MADS RUDOLPH · CV

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Engineering internship (praktik) from late January 2027, and student jobs before then. Every project below links to its write-up.

Profile

5th-semester electrical engineering student at DTU who finds faults at the bench and fixes them: a 130 µm board bow that broke PCB milling, a floating ground that would have killed a supply, a control loop identified against the wrong plant. I design and fabricate PCBs in KiCad on my own mill and laser, write firmware in C for ESP32 and AVR, and verify with a scope and tests. Available for an internship from late January 2027, with electives aimed at analog, power electronics and electroacoustics.

Projects as experience

SRM-CAM — CAM tool for a Roland SRM-20 PCB mill

Solo · in daily use at the DTU Ballerup fab lab

- Diagnosed why milled isolation traces skipped copper on a 130 µm bowed board: a 3-point routine could only fit a plane. Replaced it with a probed grid and bilinear leveling so every Z follows the real surface.
- Reverse-engineered the mill's SPI remote header with an Arduino: audited Roland's library (5 of 17 commands in use, 1 of 64 status bits read), decoded the full status word, and added a STOP that interrupts a move in flight.
- Brought double-sided registration to under 0.1 mm after the flip with a swept fit-test coupon for dowel-hole compensation; 374 commits, ~1,000 tests, Windows installer and Linux AppImage built in CI.

Vinyl ADC — discrete 3rd-order delta-sigma audio ADC

Solo · in progress, boards being milled

- Designed a stereo 48 kHz converter from TL07x op-amps, an LM311 and 74HC logic, sized to the DTU parts shop: 1.536 MHz clock, OSR 32, ~68 dB SNR in simulation.
- SPICE caught a charge pump drawn backwards (the −5 V rail simulated at +3.4 V) that every connectivity check had passed; found that comparator delay, not op-amp bandwidth, limits the clock.
- Split one unroutable single-sided board (45 wire bridges) into four milled boards with a script that proves the split matches the reference netlist.

Sub crossover — measured line-level low-pass for a Bose bass module

Solo · bring-up in progress

- Characterised the module before designing: 8.9 kΩ aux input, 0 ms latency, and a 63–203 Hz bandpass that killed the planned 50 Hz Linkwitz-Riley and moved the crossover to 85–190 Hz.
- Designed a mono-summing Sallen-Key low-pass on one TL074 with three switched corners, every part from the DTU shop; caught a missing DC bias path and a 6 V output offset in review.
- Scripted, gated bring-up on an Analog Discovery 3: three corners within 0.2 dB of the as-built model, 58 µV rms noise (instrument-limited), 3.52 V rms in before 1% THD; a fitted capacitor model predicted held-out settings to 0.7%.

Korad KD3005D — hidden UART reverse-engineered

Solo

- Found the unpopulated remote header on a bench supply and decoded its protocol with a logic analyzer: 9600 8N1, poll-only ASCII, 100–200 ms command pacing.
- Caught a floating-ground hazard with a multimeter before it destroyed hardware, and designed an isolation-safe ESP32 carrier PCB (ERC-clean, Gerbers exported for fiber-laser etching).

Skills

Bench & troubleshooting

- Oscilloscope, logic analyzer, bench PSU
- Fault diagnosis and root-cause write-ups
- Protocol reverse engineering (UART, SPI, I2C, IR)
- Calibration and measurement

PCB design & fabrication

- KiCad: schematic, footprints, layout, ERC/DRC
- Single-sided milling (Roland SRM-20), fiber-laser etching, reflow
- Gerber/Excellon, DFM for in-house processes
- IPC-2221 copper sizing

Embedded firmware

- C on ESP32 (ESP-IDF) and AVR (bare-metal ATmega)
- PID control, safety watchdogs, state machines
- Unit-tested firmware cores
- ESPHome components, Home Assistant

Analog, power & signal

- Op-amp and comparator circuits, delta-sigma modulators
- Switch-mode basics, MOSFET drive, charge pumps
- DSP: DFT/Goertzel, oversampling, decimation
- SPICE (ngspice, LTspice) and Python modelling

Tooling & AI-assisted development

- Python desktop and CLI tools (PySide6, Electron)
- Git, GitHub Actions CI, reproducible builds
- Directing AI to build software to spec, verified by tests
- MATLAB/Simulink, Fusion 360, Blender

Education

B.Eng. in Electrical Engineering (Elektroteknologi) — Sep 2024 — thesis planned autumn 2027

DTU — Technical University of Denmark, Kgs. Lyngby

115 ECTS completed. Electives in analog IC design, power electronics, electroacoustics, acoustics and EMC. Mandatory internship spring 2027, bachelor thesis autumn 2027.

STX (upper secondary), Mathematics A / Physics A — 2019 — 2022

Sønderborg Statsskole

Employment

Service and seasonal work alongside my studies. What it shows: I turn up, keep my head in a busy shift, and communicate clearly with guests and colleagues in Danish and English.

Waiter, Hotel D'Angleterre, Copenhagen — Dec 2024 — present

Bartender / waiter, Visselulles Vinbar, Sønderborg — May 2023 — Aug 2024

Line cook, Mommark Marina — Jun 2021 — Jun 2024

Production worker, JKS at Danfoss, Nordborg — Aug 2022 — Nov 2023

Ski instructor (CSIA Level 1), Sun Peaks Resort, Canada — Dec 2022 — Apr 2023

Languages

Danish (native), English (fluent).

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